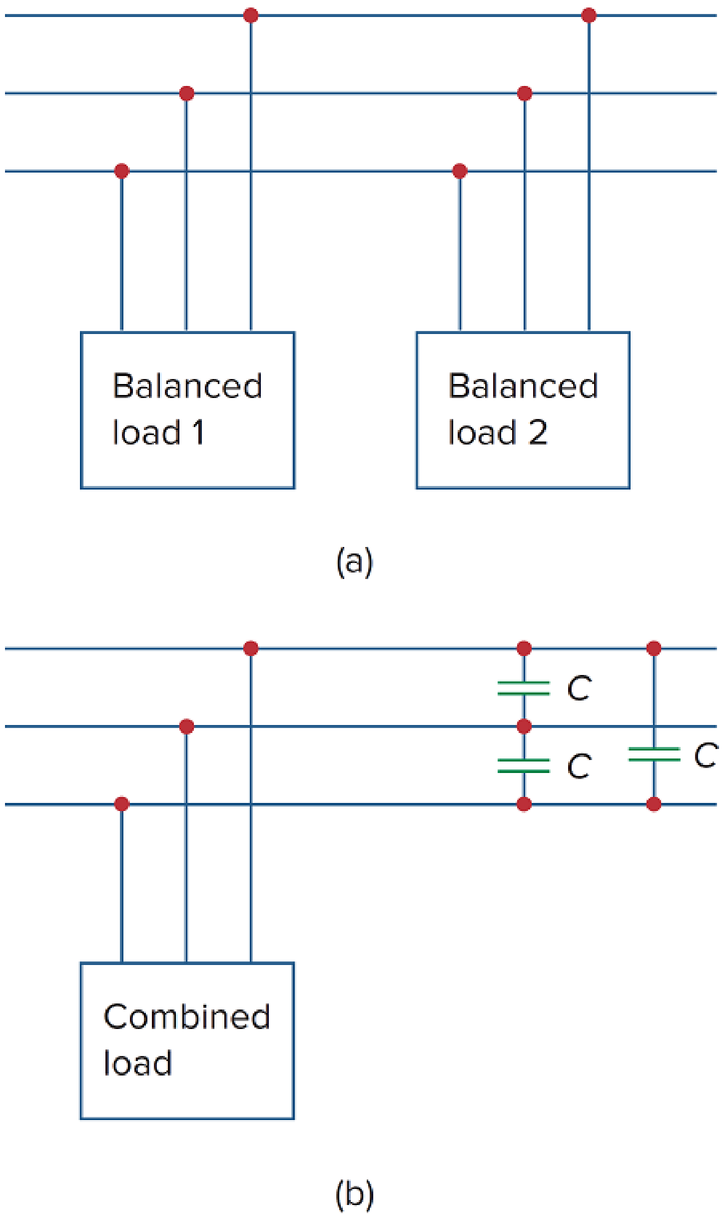
Chapter 12, Problem 8PP

Problem

Assume that the two balanced loads in Fig. 12.22(a) are supplied by an 840-V rms 60-Hz line. Load 1 is Y-connected with 30 + *j*40 Ω per phase, while load 2 is a balanced three-phase motor drawing 48 kW at a power factor of 0.8 lagging. Assuming the *abc* sequence, calculate: (a) the complex power absorbed by the combined load, (b) the kVAR rating of each of the three capacitors Δ-connected in parallel with the load to raise the power factor to unity, and (c) the current drawn from the supply at unity power factor condition.

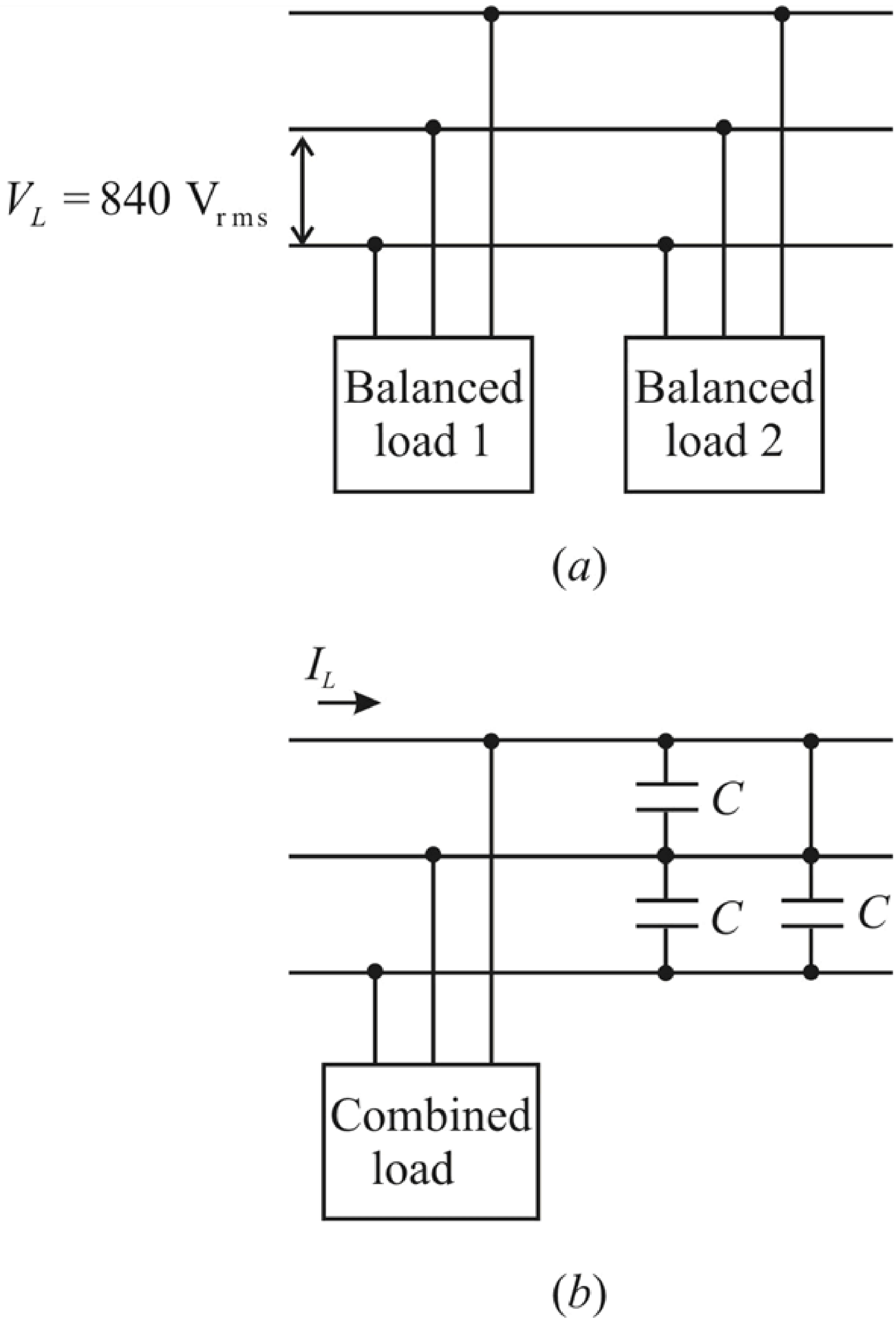
Figure 12.22



Step-by-step solution

Step 1 of 5

Given figures are



Step 2 of 5

For load

$Z = 30 + j40\Omega$

We have

$$S_1 = \frac{V_{rms}^2}{Z^*}$$

$$S_1 = \frac{(840)^2}{30 - j40}$$

$$\therefore S_1 = 8.4672 + j11.29\ \text{kVA}$$

Step 3 of 5

For load 2,

$P_2 = 48 \text{ kW}$

$\cos\theta_2 = 0.8$

$\therefore \theta_2 = 36.87°$

Hence

$Q_2 = S_2 \sin\theta_2$

$Q_2 = \left(\frac{P_2}{\cos\theta_2}\right)\sin(36.87°)$

$Q_2 = \left(\frac{48\times10^3}{0.8}\right)\times0.6$

$Q_2 = 36 \text{ kVAR}$

$\therefore S_2 = P_2 + jQ_2$

$S_2 = 48 + j36 \text{ kVA}$

Step 4 of 5

(a) Therefore the complex power absorbed by the combined load is

$S = S_1 + S_2$

$S = \left((8.4672 + j11.29) + (48 + j36)\right) \text{ kVA}$

$\therefore \boxed{S = 56.47 + j47.29 \text{ kVA}}$

(or)

$\boxed{S = 73.656\angle 39.84° \text{ kVA}}$

The real power $(P) = 56.47 \text{ kW}$

The reactive power $(Q) = 47.29 \text{ kVAR}$

Power factor $(pf) = \cos(39.94°)$

$pf = 0.7667$

Step 5 of 5

(b)

New power factor in unity

$$\cos\theta_{new} = 1$$

$$\theta_{new} = 0°$$

We have

$$Q_c = P\left(\tan\theta_{old} - \tan\theta_{new}\right)$$

$$Q_c = \left(56.47\times10^3\right)\left(\tan\left(39.94°\right) - \tan\left(0°\right)\right)$$

$$\therefore Q_c = 47.28 \text{ kVAR}$$

For each capacitor the rating is $\dfrac{47.258\times10^3}{3}$

$$= \boxed{15.76 \text{ kVR}}$$

(c) Given that

$$pf = \cos\theta = 1$$

Since $S = \dfrac{P}{\cos\theta}$

$$S = \frac{56.47\times10^3}{1}$$

$$S = 56.47 \text{ kVA}$$

Therefore

$$I_L = \frac{S}{\sqrt{3}\times V_L}$$

$$I_L = \frac{56.47\times10^3}{\sqrt{3}\times840}$$

$$\therefore \boxed{I_L = 38.813 \text{ A}}$$